# 8-2 IP Addresses and Domain Names

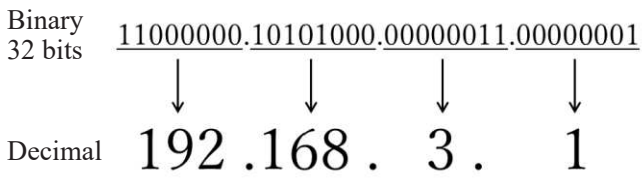
## Point!

### IP Addresses and Domain Names

(1) ([1]**IP address**) : A unique identification number equivalent to an address assigned to devices on an information and communication network.

<Features of IP Addresses>

- The method of representing an IP address using binary with ([2]**32 bits**) is referred to as ([3]**IPv4**).
- It is represented in four blocks of ([4]**8 bits**) each, separated by a dot (.) and expressed in decimal from 0 to ([5]**255**).




(2) Global IP Address and Private IP Address

[1] ([6]**Global IP address**) : An IP address used on the internet.

[2] ([7]**Private IP address**) : An IP address used within a local network, such as a LAN.

(3) Issue of IP Address Exhaustion

Due to the rapid proliferation of the internet, there are almost no new IPv4 addresses available for allocation. Therefore, we are currently transitioning to ([8]**IPv6**) IP addresses, which have been expanded to ([9]**128 bits**).

(4) ([10]**Domain name**) : A string of characters assigned to make a numerical IP address more understandable for humans.

(5) ([11]**DNS**) : A system that associates domain names with IP addresses and vice versa. The ([12]**DNS server**) is responsible for fulfilling that role.

## Warm Up

Choose the appropriate example of an IPv4 IP address for a computer from the options **A** to **D** below, and answer using the letter.

**A** 192.168.0.1
**B** kantei.go.jp
**C** 555-1234-5678
**D** https://www.example.co.jp/

### Explanation

An IPv4 address is divided into four blocks, with each block's value ranging from 0 to 255.
Therefore, **A**

## Try

Answer the following questions.

(1) Choose the appropriate example of an IPv4 IP address for a computer from the options **A** to **D** below, and answer using the letter.

**A** 00.00.11.aa.bb.cc
**B** 050-1234-5678
**C** 10.123.45.67
**D** https://www.example.co.jp/

(2) Fill in the blank in the following sentence with the correct term.

An IP address that is uniquely assigned on the internet with no duplication is referred to as a (  ) IP address.

(3) Choose the correct statement regarding IP addresses from the options **A** to **D**, and answer using the letter.

**A** The number of IP addresses in a four-byte format like 192.168.1.1 exceeds the population of the Earth, which is approximately 7 billion.
**B** The issue of IP address exhaustion is resolved by IPv6, which is represented using 128 bits.
**C** In IPv4, addresses are represented by combining four numbers ranging from 0 to 256.
**D** A global IP address can be freely used within a local network, such as within the same company.

## Exercise

**1** Answer the following questions.

(1) Choose the appropriate example of an IPv4 IP address for a computer from the options **A** to **D** below, and answer using the letter.

**A** saiyo@example.co.jp
**B** 070-1234-5678
**C** https://www.example.co.jp/
**D** 203.216.206.63

(2) Fill in the blank in the following sentence with the correct term.

An IP address that can be freely used within a local network, such as a LAN, is referred to as a (  ) IP address.

(3) Choose the correct explanation regarding DNS from the options **A** to **D** below, and answer using the letter.

**A** A unique identification number assigned to devices on an information and communication network.
**B** A string of characters assigned to an IP address to make it easier for humans to understand.
**C** A system that maps domain names to IP addresses and vice versa.
**D** An IP address that is uniquely assigned on the internet with no duplication.